Internship Report: ShadowFox Virtual Internship

Role: Data Analyst | Level: Beginner | Duration: 1 September to 30 September 2026

Name: Rudra Kumar | SF-PID: SF-PID-2026-W6251 | Institution: Amrita Vishwa Vidyapeetham, Nagercoil

1. Introduction

I joined the ShadowFox Virtual Internship as a Data Analyst. This report covers my beginner-level project. The task was to analyse a business dataset and build a spreadsheet dashboard with key metrics, trends and insights.

2. Objectives

- Clean and organise a sales dataset.
- Calculate key summary metrics.
- Study monthly, category, region and product trends.
- Present the results in charts and a simple dashboard.
- Write business-focused observations and recommendations.

3. Dataset and Tools

I used a sample retail sales dataset for 2025. It has 12 products, 3 categories (Electronics, Furniture, Stationery) and 4 regions. The data is for practice and is not real company data. I used Microsoft Excel with SUMIFS, COUNTIFS, INDEX-MATCH, charts and conditional summary tables. Work is saved on GitHub.

4. Data Cleaning

The raw data had 707 rows. After cleaning, it has 684 rows. The steps were:

- Removed 15 duplicate rows.
- Trimmed spaces and fixed letter case in 12 Region cells.
- Marked 9 missing Regions as "Unspecified" and kept the rows.
- Removed 8 rows with missing Qty, as revenue could not be found.
- Added formulas: Revenue = Qty x Price x (1 - Discount), and Profit = Revenue - (Qty x Cost).

5. Dashboard Design

The dashboard has six KPI cards: revenue, profit, profit margin, orders, average order value and units sold. It also shows the top product, top category, top region, best month, weakest month and lowest-margin product. Four charts show monthly revenue, revenue by category, revenue by region and revenue by product. All values come from formulas, so they update when the data changes.

6. Key Findings

Metric	Result
Total revenue	INR 44.6 lakh
Total profit / margin	INR 14.7 lakh / 32.9%
Orders / Avg order value	684 / INR 6,522
Best / weakest month	November (INR 5.30 lakh) / February (INR 2.62 lakh)
Q4 (Oct-Dec) share of revenue	32.3%
Top category	Furniture, 55.6% of revenue, 30.4% margin
Best-margin category	Stationery, 49.2% margin on 14.8% of revenue
Top region	South, 35.5% of revenue
Top product	Study Desk, INR 10.7 lakh
Lowest-margin product	Smart Watch, 24.0%

7. Recommendations

- Stock up and plan campaigns before October, as demand rises in Q4.

- Run an offer in February to lift the weakest month.
- Review the price or cost of the Smart Watch.
- Bundle Stationery with Furniture orders to raise profit.
- Study why East and West trail South.

8. Challenges and Learning

The main challenge was dirty data. Duplicates, mixed text case and missing values changed the totals until I fixed them. I learned to log each cleaning step, to use formulas instead of typed numbers, and to pick a chart that fits the question. I also learned to turn numbers into short, clear insights.

9. Conclusion

This project gave me hands-on practice in the full analysis flow, from raw data to a dashboard and recommendations. I plan to use these skills in my academic projects and to continue to the intermediate level.

GitHub repository: [paste your repository link here]